ModelIt!

Student Activity Pack

Why Do Animals Live in Groups?

The Power of Teamwork in Nature

Grade 3 | NGSS 3-LS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Can you name an animal that lives in a group? Why do you think they stay together?

2 Would you rather walk through a dark forest alone or with ten friends? Why?

3 Draw a group of animals working together and explain what they are doing.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Group Size | Predator Detection | Survival Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Alone

Group Size set to 1

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Small Group

Group Size set to 5

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Large Herd

Group Size set to 50

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Food Availability — How much food is nearby — larger groups need more food, which can cause competition

Warning Signal Speed — How fast the alarm call spreads through the group — faster signals save more lives

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do so many animals choose to live together in groups instead of living alone? Use evidence from your simulation to support your answers.

1 Why is 'safety in numbers' a good way to describe how animal groups work?

2 How did your model help you understand why animals are safer in groups than alone?

8 STEM Challenge: Design an Animal Alarm System

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YOUR ENGINEERING MISSION

Design a model alarm system inspired by how animal groups warn each other about predators approaching.

THE SCENARIO

A wildlife park wants to protect a group of animals from predators. Your team must design an alarm system that works like animal group behavior — the more watchers you have, the faster danger is detected. Use what you learned about group behavior to create your design!

Engineering Design Thinking

1. HOW MANY LOOKOUT STATIONS WILL YOUR ALARM SYSTEM HAVE?
2. WHAT KIND OF SIGNAL WILL THE LOOKOUTS SEND — SOUND, LIGHT, OR MOVEMENT?
3. HOW WILL YOU TEST WHETHER MORE LOOKOUTS DETECT DANGER FASTER?
4. WHAT HAPPENS IF ONE LOOKOUT MISSES THE DANGER — DOES THE SYSTEM STILL WORK?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wildlife Biologists study animals in their natural habitats, observing group behavior, migration, and survival strategies. They earn $45,000-$85,000/year.